

Riesz-energy, finiteness and every-density corollaries of the openai/math planar universal-optimality theorem, kernel-checked in Lean 4

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Abstract

The openai/math release contains an AI-generated Lean 4 proof that the triangular lattice A of covolume one minimises the lower g -energy among all locally finite planar configurations of density one, for every smooth completely monotone g (theorem `universal_energy_minimum`, Comparator-accepted by us under the pinned toolchain) [1, 2]. We derive three named consequences inside that pinned environment and check them with the Lean kernel, using only the axioms `propext`, `Classical.choice`, `Quot.sound`: (D1) for every $s > 0$ the Riesz potential $t \mapsto t^{-s/2}$ (in $t = |x - y|^2$, i.e. $|x - y|^{-s}$) is admissible, so the triangular lattice minimises the Riesz s -energy among all locally finite density-one configurations; (D1b) for $s > 2$ the triangular lattice’s Riesz s -energy (an Epstein zeta value) is finite, so D1 is a comparison with a finite, attained minimum (for $0 < s \leq 2$ the lattice side is ∞ and D1 only says $E(C) = \infty$); (D2) for every density $\rho > 0$ and every admissible g , the rescaled lattice $\rho^{-1/2}A$ has density ρ and minimises the g -energy among all locally finite density- ρ configurations; without the rescaling the statement is false (proved in Lean at $\rho = 1/4$, $s = 4$). Mathematically each step is short: complete monotonicity of $t^{-s/2}$ and a scaling argument. The upstream manuscript already states informally that its theorem covers inverse powers. What is new is the kernel-checked form of these statements, with a statement-fidelity audit, planted negative controls and re-verification by separate instances of the same AI model (producer $\neq$ verifier). Everything is conditional on the upstream theorem, which has not been peer reviewed and which, by its authors’ own account, does not claim the stronger per-potential form of the Cohn–Kumar conjecture.

1 Context and scope

Rankin, Cassels, Ennola and Diananda [5, 6, 7, 8, 9] showed that the triangular lattice minimises the Epstein zeta function $\sum_{a \neq 0} |a|^{-s}$, $s > 2$, among planar lattices of fixed covolume, and Montgomery [10] did the same for Gaussian (theta) sums. Comparison with *all* configurations is the planar case of the universal-optimality question of Cohn and Kumar [3, Conj. 9.4], proved in dimensions 8 and 24 by Cohn, Kumar, Miller, Radchenko and Viazovska [4]. Partial planar results concern restricted competitor classes [11] and small perturbations [12].

On 2026-09-23 the openai/math release posted a manuscript [1] claiming the energy-comparison conclusion in the plane, together with a Lean proof of it. By its own account that manuscript constructs sharp auxiliary functions for every Gaussian but not for each mixed completely monotone potential, and it does not claim that stronger per-potential form of the Cohn–Kumar conjecture. The release of 2026-10-07 withdrew three unrelated manuscripts (on Weil classes of abelian eightfolds, Kuga–Satake for K3, and the Hodge conjecture for products of K3 surfaces). It did not touch the triangular manuscripts, and the Lean files we use are byte-identical at its commit (Section 2).

What this note is and is not. It is a kernel-checked derivation, inside upstream’s pinned environment, of three statements physicists use: Riesz energies, their finiteness, and arbitrary density. It is *not* an independent proof of universal optimality, *not* a new mathematical idea (the manuscript already notes that inverse powers are included), and *not* a peer review of the upstream proof. If the upstream statement is a faithful formalisation and its proof is correct, then D1–D2 are theorems; otherwise they are only implications.

2 The upstream statement and a fidelity audit

Let $\text{Plane} = \mathbb{R}^2$ with the Euclidean norm, and let $N_R(C)$ be the number of points of C in the closed disk of radius R about 0. The upstream Lean file (`ComparatorChallenges/TriangularEnergy.lean`, sha256 `af51979c...0616`) defines:

- g is *admissible* if $g \in C^\infty(0, \infty)$, $g \geq 0$, and $(-1)^r g^{(r)}(t) \geq 0$ for all $r \geq 0$ and $t > 0$;
- C has *density one* if $N_R(C)/(\pi R^2) \rightarrow 1$;
- $E_g(C) = \liminf_{R \rightarrow \infty} N_R^{-1} \sum_x \sum_{y \neq x} g(|x - y|^2)$, the sums running over points of C in the closed R -disk, with values in $[0, \infty]$;
- $A = \{b^{-1/2}(j + k/2, kb) : j, k \in \mathbb{Z}\}$, $b = \sqrt{3}/2$ (covolume one), and $E_g^{\text{lat}} = \sum_{a \in A \setminus 0} g(|a|^2) \in [0, \infty]$.

The theorem is: for admissible g and locally finite C of density one, $E_g^{\text{lat}} \leq E_g(C)$ and $E_g^{\text{lat}} = E_g(A)$. The challenge file states it with `sorry` by design; the proof is in `OAI/Analysis/Triangular/Energy/Universal.lean`, and `Comparator` certifies that the two statements agree. The file has the same sha256 at upstream commits `adc7f124` and `fd4aeeb2`, and `git diff` between them touches no file under `lean/OAI/Analysis/Triangular/`, `LatticeSummability.lean`, the toolchain or the lake manifest.

We compared each item with the lower-energy and centred-density formulation of [4, Defs. 1.1–1.3] that the manuscript cites (`results/discovery/statement_fidelity.md`). They match, including the ordered-pair normalisation by N_R and the convention that divergent sums equal ∞ . Two points remain. First, upstream uses closed disks and we did not analyse whether the open-disk lower energy can differ. Second, the audit was done by the producing agent; the independent verifiers read the same definitions separately.

3 Results

All statements below are Lean theorems in the AutoevolveAI repository (`formal_cert/Triangular/`), compiled against upstream’s pinned Lean 4.34.1 / Mathlib `d13f23b7` / openai/math build. Here $\text{riesz}_s(t) = t^{-s/2}$, so $\text{riesz}_s(|x - y|^2) = |x - y|^{-s}$; this identity is kernel-checked for $|x - y| > 0$ (`TriangularControls.riesz_sq`).

Theorem 1 (`D1`, `TriangularRiesz.riesz_admissible`, `triangular_riesz_optimal`). *For every $s > 0$, riesz_s is admissible. Hence for every locally finite C of density one, $\sum_{a \in A \setminus 0} |a|^{-s} \leq E_{\text{riesz}_s}(C)$.*

Proof idea. Mathlib’s `Real.iter_deriv_rpow_const` gives $\frac{d^r}{dt^r} t^p = \text{desc}_r(p) t^{p-r}$. With $p = -s/2$, $\text{desc}_r(-s/2) = (-1)^r \text{asc}_r(s/2)$ (`ascPochhammer_eval_neg_eq_descPochhammer`), and $\text{asc}_r(s/2) > 0$, which gives the sign pattern. Smoothness away from 0 is `Real.contDiffAt_rpow_const_of_ne`. No value at $t \leq 0$ is ever used.

Theorem 2 (D1b, `TriangularRiesz.latticeEnergy_riesz_lt_top`). For every $s > 2$, $\sum_{a \in A \setminus 0} |a|^{-s} < \infty$.

Proof idea. `norm_sq_triangularPoint`: $|p(j, k)|^2 = \frac{2}{\sqrt{3}}(j^2 + jk + k^2)$. Reindex the sum over $A \setminus 0$ by the injective parametrisation, map $k \mapsto -k$ to match the form $j^2 - jk + k^2$, and apply the `openai/math` lemma `LatticeSummability.summable_normForm_neg_rpow` with $\sigma = s/2 > 1$. For $0 < s \leq 2$ the sum diverges (standard; not kernel-checked; the verifier confirmed $2\pi \log R$ growth numerically at $s = 2$), so D1 compares with a finite minimum exactly for $s > 2$.

For $\rho > 0$ say that C has *density* ρ if $N_R(C)/(\pi R^2) \rightarrow \rho$. At $\rho = 1$ this is upstream’s definition by `Iff.rfl`.

Theorem 3 (D2; `universal_any_density`, `triangular_optimal_any_density`, `attained`). Let g be admissible and $\rho > 0$. Then $\rho^{-1/2}A$ is locally finite and has density ρ , and for every locally finite C of density ρ ,

$$E_g(\rho^{-1/2}A) = \sum_{a \in A \setminus 0} g(|a|^2/\rho) \leq E_g(C).$$

Proof idea. $t \mapsto g(ct)$ is admissible for $c > 0$; the identity $\frac{dr}{dt}g(ct) = c^r g^{(r)}(ct)$ holds globally from `deriv_comp_mul_left` with no differentiability hypothesis. Next, $N_R(\lambda C) = N_{R/\lambda}(C)$, so $\sqrt{\rho}C$ has density one, and the disk energies transform as $E_{g,R}(\lambda C) = E_{g(\lambda^2 \cdot), R/\lambda}(C)$. The $\liminf$ is invariant under $R \mapsto R/\lambda$, because this order isomorphism maps `atTop` to itself. Applying upstream’s theorem to $g(\cdot/\rho)$ and $\sqrt{\rho}C$ gives the inequality, and its attainment conjunct gives the equality. The density and attainment lemmas (`scaled_density`, `attained`) were written by the independent D2 verifier and merged with attribution.

Corollary 4 (`riesz_any_density_homogeneous`). For $s > 0$, $\rho > 0$ and locally finite C of density ρ : $\rho^{s/2} \sum_{a \in A \setminus 0} |a|^{-s} \leq E_{\text{riesz}_s}(C)$ in $[0, \infty]$. The homogeneity step $\sum_a \text{riesz}_s(\rho^{-1}|a|^2) = \rho^{s/2} \sum_a |a|^{-s}$ is kernel-checked (`riesz_lattice_homogeneous`). Informative for $s > 2$.

Negative control. The unrescaled claim “ $E_g^{\text{lat}} \leq E_g(C)$ for every locally finite C of density ρ ” is false at $\rho = 1/4$, $s = 4$, and this is proved in Lean (`TriangularDensity.unscaled_false`). The witness $C = 2A$ has energy $\sum (4|a|^2)^{-2}$, which is strictly smaller, and finite by D1b. So the ρ^{-1} in D2 cannot be dropped. For $\rho \geq 1$ the unrescaled claim is true but weaker (it follows from D2 because admissible g are non-increasing; not kernel-checked). A further control: $t \mapsto t$ ($s = -2$) is not admissible (`riesz_neg_two_not_admissible`).

4 Verification

- **Kernel and axioms.** `Riesz.lean`, `RieszFinite.lean`, `AnyDensity.lean` and `Controls.lean` compile with exit code 0 (each about 13–20s once upstream’s `olean`s are cached; two harmless deprecation warnings). `#print axioms` reports `{propext, Classical.choice, Quot.sound}` for every theorem above and for upstream’s `universal_energy_minimum`. A planted `sorry` and a planted axiom in the same audit file are both flagged (`sorryAx`, `ctl_cheat`). There is no `sorry`, `axiom` or `native_decide` in the four proof files; `Axioms.lean` contains only the two planted controls.
- **Preregistration.** D1b and D2 had preregistrations with kill criteria committed before their Lean files (3–4 minutes earlier, per git log; commit order is the only evidence of timing;

results/discovery/D1b,D2/preregistration.json). D1 was not preregistered: the proposal document (docs/DISCOVERY_PROPOSAL_2026-10-10.md), which assigned it a prior of about 85%, was committed together with its Lean file.

- **Re-verification.** D1/D1b and D2 were each re-verified by a separate instance of the same AI model (producer $\neq$ verifier), which recompiled the files and wrote its own axiom audit, controls and numerics. Both verdicts were CONFIRMED_WITH_CORRECTIONS. The corrections were wording and scope, plus (D2) adding the verifier’s lemmas `scaled_density`, `attained` and `unscaled_false`, which no second party wrote or reviewed line by line. The final `AnyDensity.lean` and `Controls.lean` were recompiled with clean axioms by the producer and by a pre-publication referee (another separate instance), but not re-examined by the D2 verifier. The verifiers’ controls also included a check that the D1b proof cannot pass its only s -dependent step at $s = 2$ (scratch file, not shipped).
- **Numerics** (verifier code, `mpmath`; for the Riesz rows, closed forms agree with direct sums to about 10^{-7} ; the referee reproduced every row independently). Table 1: the triangular lattice beats the square lattice at equal density in every case, and for Riesz $s = 4$ the ratio 0.959605 does not depend on ρ , as homogeneity predicts.

$g(t)$	ρ	triangular	square	square – triangular
$t^{-3/2}$ ($s = 3$)	1	8.8927451004	9.0336216831	0.1409
t^{-2} ($s = 4$)	1	5.7833592997	6.0268120397	0.2435
t^{-3} ($s = 6$)	1	4.1412565472	4.6589136156	0.5177
e^{-t}	0.5	0.60238788592	0.61630926604	1.39×10^{-2}
e^{-t}	2	5.2831853119385	5.2831853744169	6.25×10^{-8}
e^{-t}	4	11.566370614359172955	11.566370614359173314	3.59×10^{-16}
t^{-2} ($s = 4$)	0.5	1.445835	1.506699	ratio 0.959605
t^{-2} ($s = 4$)	2	23.13337	24.10718	ratio 0.959605
t^{-2} ($s = 4$)	4	92.53347	96.42874	ratio 0.959605

Table 1: Per-point lattice energies $\sum_{y \in L \setminus \{0\}} g(|y|^2)$ at density ρ , with $L = \rho^{-1/2}L_1$ and L_1 the covolume-one triangular or square lattice (verifier numerics, reproduced by the referee). At $\rho = 4$ the Gaussian gap is below double precision, and a float64 run misreports the order; the 60-digit run resolves it.

5 Limitations

1. Everything is conditional on upstream’s `universal_energy_minimum`. That proof is AI-generated and has not been peer reviewed, though we checked that it is accepted by Comparator, with clean axioms, under the pinned toolchain. Its statement has been audited by AI agents, not by a human expert.
2. The Cohn–Kumar conjecture in its per-potential form (a sharp auxiliary function for each potential) is not established, upstream or here.
3. For $0 < s \leq 2$, D1 only says $E(C) = \infty$; renormalised logarithmic ($s = 0$) and Riesz ($0 < s < 2$) energies are treated in the upstream manuscript, not here; neither treats $s = 2$.
4. Closed versus open disks in the energy definition was not analysed.

5. Other admissible families (Yukawa, $e^{-\kappa\sqrt{t}}/\sqrt{t}$) were not attempted. They need complete monotonicity of compositions, which current Mathlib does not offer in usable form.

Reproducibility. Lean sources: `Riesz.lean`, `RieszFinite.lean`, `AnyDensity.lean`, `Controls.lean` and `Axioms.lean` in `formal_cert/Triangular/` (sha256 in the bundle’s `MANIFEST.json`). Upstream: `openai/math` commit `adc7f124`, unchanged at `fd4aeeb2` for the files used. Toolchain: Lean 4.34.1, Mathlib `d13f23b7`. Compiler driver: `scripts/certified_numerics/lean_pinned.py`. Verification reports and numerics scripts: `results/discovery/`. Repository: <https://github.com/xaviercallens/AutoevolveAI>, Lean sources at commit `9eb1773f0e7b2bc394d4b685df29cf1f88327f49`. Method: SocrateAI-Scientific-Elenchus (tiered hypotheses with predictions and kill criteria) [13].

AI use. The Lean proofs, audit and text were produced by AI agents (Claude) under the author’s direction; the verification runs were separate agent instances. The author takes responsibility for the claims as scoped above.

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